Functions, Modulus and Quadratics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

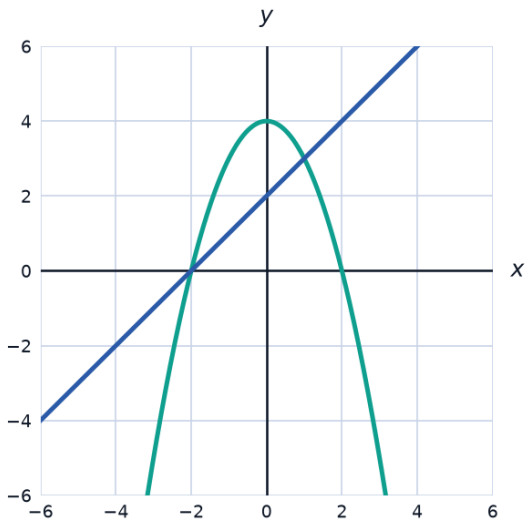
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

2 [Foundational] • Target: Function Mapping, Domain/Range and Inverses

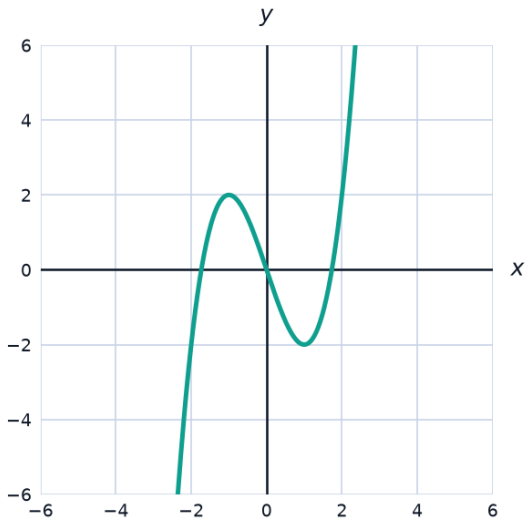
The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.



© ClipSAT — clipsat.org | All Rights Reserved

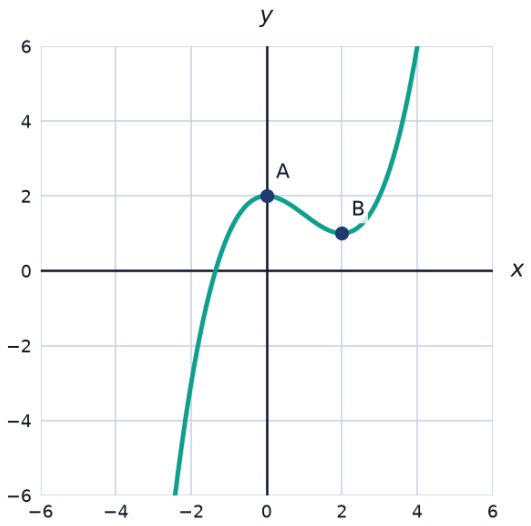
3 [Foundational] • Target: Modulus Equations and Inequalities

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.



4 [Foundational] • Target: Quadratic Discriminant and Line-Parabola Intersections

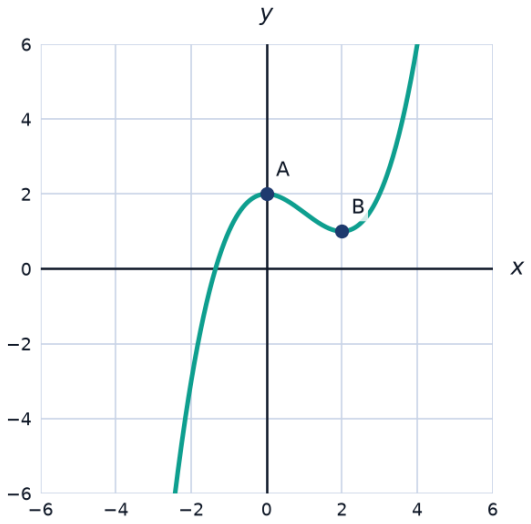
A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.



© ClipSAT — clipsat.org | All Rights Reserved

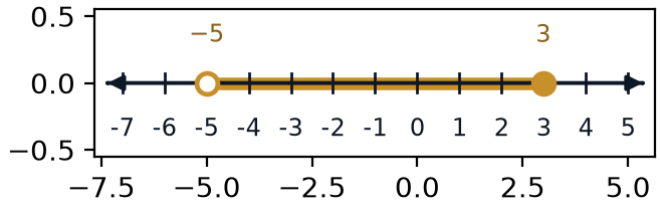
5 [Foundational] • Target: Completing the Square and Applied Optimization

Find the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.



6 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The variables x and y satisfy the relationship $y = Ax^n$, where A and n are constants. When $\log_2 y$ is plotted against $\log_2 x$, a straight line is obtained which passes through the points $(1, 5)$ and $(4, -1)$. What is the value of y when $x = 4$?



7 [Foundational] • Target: Function Mapping, Domain/Range and Inverses

Variables x and y satisfy the equation $y = Ax^k$, where A and k are constants, with $x > 0$. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$. Find the value of x when $y = 18$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modulus Equations and Inequalities

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

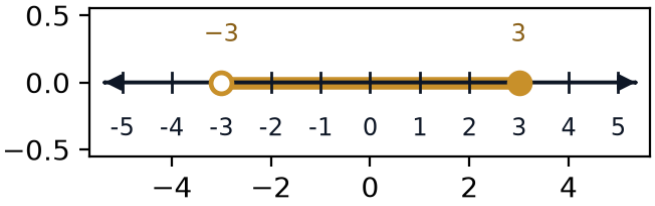
© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant. Given that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

10 [Intermediate] • Target: Completing the Square and Applied Optimization

The quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.



11 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The variables x and y satisfy the equation $y = Ab^x$, where A and b are positive constants. When $\log_2 y$ is plotted against x, a straight line is obtained that passes through the points $(2, 7)$ and $(5, 13)$. Find the value of y when $x = 4$.

12 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?

13 [Intermediate] • Target: Modulus Equations and Inequalities

The function f is defined for $x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

14 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

The function f is defined for all real numbers x by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of f?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

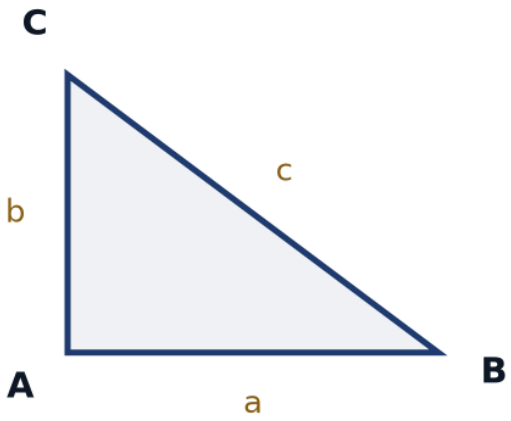
The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

17 [Challenging] • Target: Function Mapping, Domain/Range and Inverses • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A particle moves in a straight line with acceleration $a(t) = 6t - 12$ m/s^2. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?

18 [Challenging] • Target: Modulus Equations and Inequalities • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

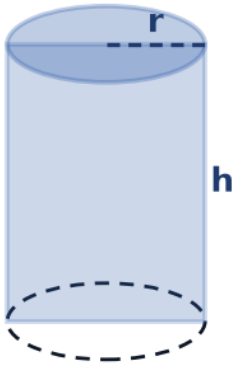
In triangle ABC, the side lengths are $AB = (x + 2)$ cm, $BC = (3x - 1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.



© ClipSAT — clipsat.org | All Rights Reserved

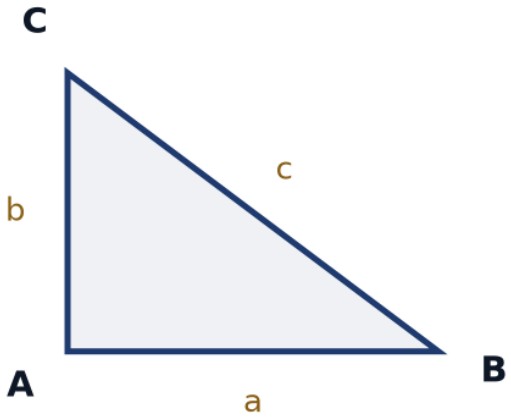
19 [Challenging] • Target: Quadratic Discriminant and Line-Parabola Intersections • Cross-Lesson: 6.01 Permutations and Combinations

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm.



20 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 7.01 Differentiation and Rates of Change

Two parallel lines, L_1 and L_2, have n and $n + 2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$. Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.



© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|2x + 3| = 7$.

a. $x = 3$ and $x = -3$

b. $x = -2$ and $x = -\frac{10}{2}$

c. $x = 2$ only

d. $x = 2$ and $x = -\frac{10}{2}$

22 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|3x + 3| = 9$.

a. $x = 2$ and $x = -\frac{12}{3}$

b. $x = 3$ and $x = -3$

c. $x = -2$ and $x = -\frac{12}{3}$

d. $x = 2$ only

23 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|4x + 3| = 11$.

a. $x = 2$ only

b. $x = 3$ and $x = -3$

c. $x = 2$ and $x = -\frac{14}{4}$

d. $x = -2$ and $x = -\frac{14}{4}$

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|5x + 3| = 13$.

a. $x = 2$ and $x = -\frac{16}{5}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{16}{5}$

d. $x = 3$ and $x = -3$

25 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|6x + 3| = 15$.

a. $x = 3$ and $x = -3$

b. $x = 2$ and $x = -\frac{18}{6}$

c. $x = 2$ only

d. $x = -2$ and $x = -\frac{18}{6}$

26 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. $x = 2$ and $x = -\frac{20}{7}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|8x + 3| = 19 \).

a. \(x = 2 \) and \(x = -\frac{22}{8} \)

b. \(x = 2 \) only

c. \(x = -2 \) and \(x = -\frac{22}{8} \)

d. \(x = 3 \) and \(x = -3 \)

28 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|9x + 3| = 21 \).

a. \(x = 2 \) only

b. \(x = 2 \) and \(x = -\frac{24}{9} \)

c. \(x = 3 \) and \(x = -3 \)

d. \(x = -2 \) and \(x = -\frac{24}{9} \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

The functions f and g are defined for $x \in \mathbf{R}$ by $f(x) = 2x - 5$ and $g(x) = x^2 + 3$. Find the composite value $(g \circ f)(4)$.

a. 12

b. 21

c. 14

d. 9

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

A function is defined by $f(x) = \frac{3x+1}{x-2}$ for $x \neq 2$. Find an expression for the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$

b. $f^{-1}(x) = \frac{x-2}{3x+1}, \quad x \neq -\frac{1}{3}$

c. $f^{-1}(x) = \frac{2x-1}{x+3}, \quad x \neq -3$

d. $f^{-1}(x) = \frac{3x-2}{x-1}, \quad x \neq 1$

31 [Intermediate] • Target: Function Mapping, Domain/Range and Inverses

The square root function $h(x) = 4 - \sqrt{2x-6}$ is defined on its maximal real domain. Determine the domain and range of h.

a. Domain: $x \geq 3$, Range: $h(x) \leq 4$

b. Domain: $x \geq 3$, Range: $h(x) \geq 4$

c. Domain: $x > 3$, Range: $h(x) \in \mathbf{R}$

d. Domain: $x \leq 3$, Range: $h(x) \leq 0$

32 [Intermediate] • Target: Modulus Equations and Inequalities

Solve the linear modulus equation $|3x - 4| = 2x + 1$ for all valid real solutions.

a. $x = 5$ and $x = \frac{3}{5}$

b. $x = 5$ only

c. $x = -1$ and $x = 3$

d. No real solutions

33 [Intermediate] • Target: Modulus Equations and Inequalities

Find the complete solution set for the absolute value inequality $|2x - 7| < 5$.

a. $1 < x < 6$

b. $x < 1$ or $x > 6$

c. $-6 < x < 1$

d. $2 < x < 5$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Modulus Equations and Inequalities

Solve the two-modulus equation $|x+2| = |2x-5|$.

a. $x = 7$ and $x = 1$

b. $x = 7$ and $x = -1$

c. $x = 3$ and $x = -3$

d. $x = \frac{7}{3}$ only

35 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

Find the set of values of k for which the quadratic equation $x^2 + (k+2)x + (2k+1) = 0$ has two distinct real roots.

a. $k < 0$ or $k > 4$

b. $0 < k < 4$

c. $k \leq 0$ or $k \geq 4$

d. $-4 < k < 0$

36 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

The line $y = 2x + c$ is tangent to the parabola $y = x^2 - 4x + 7$. Determine the exact value of the constant c.

a. $c = -2$

b. $c = 2$

c. $c = -7$

d. $c = 3$

37 [Intermediate] • Target: Quadratic Discriminant and Line-Parabola Intersections

Find the range of values of m for which the line $y = mx - 1$ does not intersect the curve $y = 2x^2 + 3x + 1$.

a. $-1 < m < 7$

b. $m < -1$ or $m > 7$

c. $-7 < m < 1$

d. $m < 3$ or $m > 5$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

By completing the square, express $f(x) = 2x^2 - 12x + 23$ in the form $a(x-h)^2 + k$, and state the minimum value of $f(x)$.

a. $2(x-3)^2 + 5$, minimum value 5

b. $2(x-3)^2 + 23$, minimum value 23

c. $2(x-6)^2 - 13$, minimum value -13

d. $(2x-6)^2 + 5$, minimum value 5

39 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 6.01 Permutations and Combinations

A rectangular field has three sides fenced with 80 m of fencing against a river. Find the maximum possible area.

a. $800\ \text{m}^2$

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

40 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find all real solutions to the biquadratic equation $x^4 - 10x^2 + 9 = 0$.

a. $x = \pm 1$ and $x = \pm 3$

b. $x = 1$ and $x = 9$

c. $x = \pm 1$ only

d. $x = \pm\sqrt{10}$

41 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The function $f(x) = (x+11)^2 - 4$ has domain $x \geq -11$. State the range of f.

a. $f(x) \leq -4$

b. $f(x) \geq -4$

c. $f(x) \geq -11$

d. $x \geq -11$

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Function Mapping, Domain/Range and Inverses • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

The function $f(x) = (x + 8)^2 - 10$ has domain $x \geq -8$. State the range of f.

a. $f(x) \geq -8$

b. $f(x) \geq -10$

c. $f(x) \leq -10$

d. $x \geq -8$

43 [Challenging] • Target: Modulus Equations and Inequalities • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The function $f(x) = (x + 7)^2 - 15$ has domain $x \geq -7$. State the range of f.

a. $f(x) \geq -7$

b. $x \geq -7$

c. $f(x) \leq -15$

d. $f(x) \geq -15$

44 [Challenging] • Target: Quadratic Discriminant and Line-Parabola Intersections • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The function $f(x) = (x + 11)^2 + 5$ has domain $x \geq -11$. State the range of f.

a. $x \geq -11$

b. $f(x) \geq -11$

c. $f(x) \leq 5$

d. $f(x) \geq 5$

45 [Challenging] • Target: Completing the Square and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$

c. $f(x) \geq -6$

d. $x \geq -6$

© ClipSAT — clipsat.org | All Rights Reserved